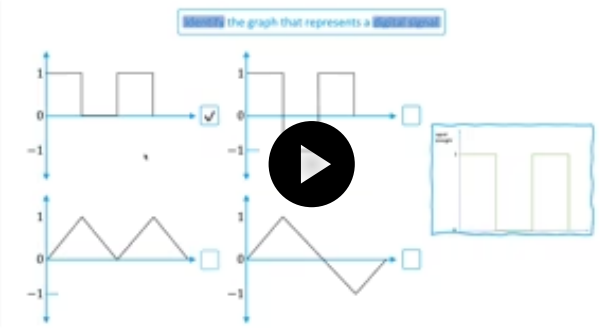
Digital & Analogue Signals

Exam code: 0625 & 0972

Download PDF

Written by: Katie M
Reviewed by: Caroline Carroll
Updated on 1 October 2024



Did this video help you? Yes No

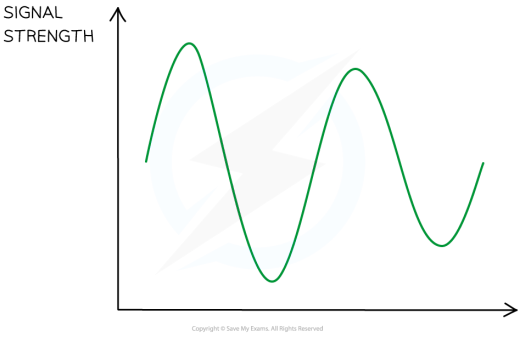
Help me understand

Digital & analogue signals

Extended tier only

- There are two types of signals:
  - Analogue
  - Digital
- **Analogue signals vary continuously** - they can take any value

An analogue signal

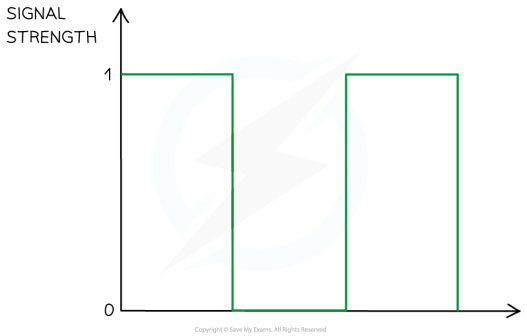


An analogue signal is continuously varying, taking any value

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **High**s and **low**s, or
- **On**s and **off**s

A digital signal



A digital signal can only take one of two values – 0 or 1

Transmission of sound

Extended tier only

- Sound waves that can be transmitted as a **digital** or **analogue** signal

- Signals for speech or music are made up of varying frequencies
  - To make out the information clearly, the signal needs to be transmitted with as little interference as possible
- The signal undergoes **digital sampling**
- It is converted **both before transmission** and **after being received**
  - Before transmission: the signal is converted from **analogue** to **digital**
  - After being received: the signal is converted back from **digital** to **analogue**

Benefits of digital signalling

Extended tier only

- The benefits of digitally sampling a signal include:
  - **Increased rate** of data transmission
  - **Increased range** due to accurate signal regeneration

Digital sampling an analogue signal

Analogue v digital signal

- The key **advantages** of data transmission in **digital** form compared to analogue are:
  - The signal can be **regenerated** so there is minimal **noise**
  - Due to accurate signal regeneration, the **range** of digital signals is **larger** than the range of analogue signals (they can **cover larger distances**)
  - Digital signals enable an **increased rate of transmission of data** compared to analogue
  - Extra data can be added so that the signal can be checked for errors

Examiner Tips and Tricks

Digital sampling is when an analogue signal is converted into a digital signal by recording sound at set time intervals.

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.